

Experiment 6

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Aim

To understand the concept and working of cursors in PL/SQL for row-by-row data processing, and to analyse how implicit cursors, explicit cursors, and cursor attributes are used to implement business logic on multiple rows in a database table.

Software Requirements

- Database Management System:
 - Oracle Database Express Edition (Oracle XE)
 - PostgreSQL Database
- Database Administration Tool / Client Tool:
 - Oracle SQL Developer (for Oracle XE)
 - pgAdmin (for PostgreSQL)

Objectives

- To implement and analyse the use of implicit cursors, explicit cursors, and cursor attributes for processing multiple rows from a database table and applying business logic effectively.

Problem Statement

- In real-world enterprise applications, database queries often return multiple rows that need to be processed individually to apply specific business rules.

Practical/Experiment Steps

- Schema Provisioning: Created the Employees table with specific data types for IDs, names, roles, and salaries to simulate a corporate database environment.
- Implicit Cursor Analysis: Implemented an anonymous PL/SQL block to perform a single-row UPDATE operation, utilizing the internal SQL cursor to monitor DML success.
- Explicit Cursor Design: Defined a named cursor (c_engg) to fetch specific records based on the 'Engineer' role, enabling structured row-by-row processing.
- Attribute Integration: Incorporated cursor attributes such as %FOUND, %NOTFOUND, %ROWCOUNT, and %ISOPEN to programmatically control the execution flow.
- Business Logic Application: Developed logic to apply salary increments and format record-specific output, demonstrating how enterprise rules are applied to individual data rows.

Procedure

- Logged into the database environment and ensured SERVEROUTPUT was enabled to capture procedural messages.
- Executed the CREATE TABLE and INSERT scripts to establish the foundational employee dataset.
- Initiated a PL/SQL block using an implicit cursor to update a specific employee's salary and verified the change using SQL%ROWCOUNT.
- Declared an explicit cursor in a separate block, specifically targeting employees with the 'Engineer' role.
- Opened the explicit cursor and used the %ISOPEN attribute to confirm the cursor was successfully initialized in memory.
- Implemented a LOOP structure to sequentially FETCH data from the active set into local variables (v_name, v_role).
- Used the %NOTFOUND attribute as an exit condition to terminate the loop once all relevant records were processed.
- Monitored the %ROWCOUNT attribute to display a dynamic count for each record retrieved during the iteration.
- Closed the cursor explicitly to release system resources and verified the closure using the NOT c_engg%ISOPEN condition.

Input/Output Analysis

SQL Input Queries

```
CREATE TABLE Employees(  
    emp_id NUMBER PRIMARY KEY,  
    name VARCHAR(20),  
    role VARCHAR(20),  
    salary NUMBER  
);
```

Output

Query result	Script output	DBMS output	Explain Plan	SQL history
 				
<pre>SQL> CREATE TABLE Employees(emp_id NUMBER PRIMARY KEY, name VARCHAR(20), role VARCHAR(20),... Show more...</pre>				
Table EMPLOYEES created. Elapsed: 00:00:00.013				

SQL Input Queries

```
INSERT INTO Employees VALUES (101, 'Mary', 'Engineer', 85000);  
INSERT INTO Employees VALUES (102, 'Amit', 'Analyst', 45000);  
INSERT INTO Employees VALUES (103, 'Sara', 'Engineer', 90000);  
INSERT INTO Employees VALUES (104, 'Sam', 'Analyst', 80000);  
COMMIT;
```

Output

Query result	Script output	DBMS output	Explain Plan	SQL history
 				
<pre>SQL> INSERT INTO Employees VALUES (101, 'Mary', 'Engineer', 85000)</pre>				
1 row inserted. Elapsed: 00:00:00.015				
<pre>SQL> INSERT INTO Employees VALUES (102, 'Amit', 'Analyst', 45000)</pre>				
1 row inserted. Elapsed: 00:00:00.002				
<pre>SQL> INSERT INTO Employees VALUES (103, 'Sara', 'Engineer', 90000)</pre>				
1 row inserted. Elapsed: 00:00:00.001				
<pre>SQL> INSERT INTO Employees VALUES (104, 'Sam', 'Analyst', 80000)</pre>				
1 row inserted. Elapsed: 00:00:00.001				

SQL Input Queries

```
SET SERVEROUTPUT ON;
```

```
DECLARE
```

```
    v_target_id NUMBER :=101;
```

```
BEGIN
```

```
    UPDATE Employees
```

```
    SET salary = salary*1.10
```

```
    WHERE emp_id = v_target_id;
```

```
IF SQL%FOUND THEN
```

```
    DBMS_OUTPUT.PUT_LINE('ATTRIBUTE SQL%FOUND TRUE');
```

```
    DBMS_OUTPUT.PUT_LINE('ROW(S) UPDATED '||SQL%ROWCOUNT);
```

```
ELSE
```

```
    DBMS_OUTPUT.PUT_LINE('NO ROW UPDATED');
```

```
END IF;
```

```
END;
```

Output



SQL Input Queries

```
DECLARE
```

```
    CURSOR c_engg IS SELECT name, role FROM Employees WHERE role =
    'Engineer';
```

```
    v_name Employees.name%TYPE;
```

```
    v_role Employees.role%TYPE;
```

```
BEGIN
```

```
    OPEN c_engg;
```

```
IF c_engg%ISOPEN THEN
```

```
    DBMS_OUTPUT.PUT_LINE('CURSOR IS OPEN');
```

```

END IF;

LOOP
    FETCH c_engg INTO v_name, v_role;

    IF c_engg%NOTFOUND THEN
        DBMS_OUTPUT.PUT_LINE('%NOTFOUND IS TRUE, NO MORE
RECORDS');
        EXIT;
    END IF;

    DBMS_OUTPUT.PUT_LINE('ENGINEER # '||c_engg%ROWCOUNT|| ': '
|| v_name || ' ' || v_role);

END LOOP;

CLOSE c_engg;

IF NOT c_engg%ISOPEN THEN
    DBMS_OUTPUT.PUT_LINE('CURSOR CLOSED SUCCESSFULLY');
END IF;
END;

```

Output

Query result	Script output	DBMS output	Explain Plan	SQL history
  <pre> CURSOR IS OPEN ENGINEER # 1: Mary Engineer ENGINEER # 2: Sara Engineer NOTFOUND IS TRUE, NO MORE RECORDS CURSOR CLOSED SUCCESSFULLY PL/SQL procedure successfully completed. Elapsed: 00:00:00.112 </pre>				

Learning Outcomes

- Gained proficiency in differentiating between implicit cursors for DML and explicit cursors for multi-row queries.
- Learnt using %FOUND, %NOTFOUND, and %ROWCOUNT to manage loop iterations and verify operation success.
- Understanding the importance of the OPEN, FETCH, and CLOSE lifecycle to ensure efficient memory management.
- Ability to develop PL/SQL programs that apply complex business logic to individual records within a larger dataset.